



EMERITUS

Learn. From the world's best.

Masterclass 1: Analysing Your Daily Routine



Unlocking Insights Through Data



What This Masterclass is About?



Introduction to fundamental statistical concepts.



Hands-on analysis of personal daily routines.

What This Masterclass is About?



Learn to collect, clean, and interpret data using Google Sheets.



Discover how small, data-driven decisions can improve daily life.

Overview



Learning Objectives

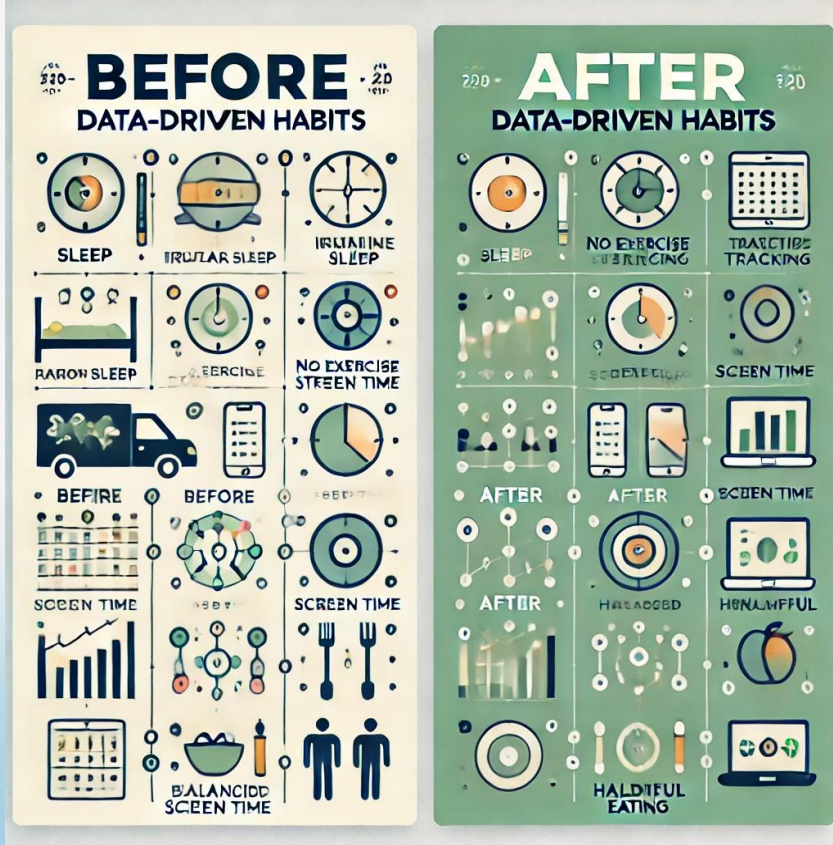
Understand key statistical terms: Mean, Median, Mode, Correlation.

Develop skills in collecting and organizing personal data.

Use Google Sheets for basic data analysis and visualization.

Identify patterns and make informed decisions based on data insights.

Why Analyze Your Daily Routine?



Small changes in habits can lead to big improvements in health and productivity.

Data helps you understand patterns and make informed decisions.

Example: Tracking sleep can reveal its impact on mood and energy levels.

What We'll Track?

Hours of Sleep

Steps Walked

Screen Time (hours)

Water Intake (liters)

Mood Rating (1-10)

Key Analysis Questions



What is the average number of hours slept over the week?

What is the total number of steps walked?

How does sleep duration correlate with mood rating?

How does screen time fluctuate with changes in sleep?

Key Analysis Questions



Which day had the **highest** and **lowest** screen time?

Is there a connection between **steps walked** and **mood rating**?

How does Water Intake correlate with mood rating?

Mean

The **mean** (also called the **average**) is a measure of central tendency that represents the typical value in a dataset. It is calculated by:

$$\text{Mean} = \frac{\sum \text{All Values}}{\text{Number of Values}}$$

Example:

Consider the numbers: **4, 7, 10, 6, 8**

$$\text{Mean} = \frac{4 + 7 + 10 + 6 + 8}{5} = \frac{35}{5} = 7$$

The mean of this dataset is **7**.

When to Use the Mean and when not ?

- ✓ Best used when data is **evenly distributed** (without extreme outliers).
- ✗ If the dataset has **outliers** (e.g., 100 in a dataset of small numbers), the mean can be misleading.

Median

The **median** is the middle value of a dataset when arranged in **ascending** or **descending** order. It is a measure of central tendency that helps identify the center of the data.

How to Calculate the Median?

1. **Arrange** the numbers in order.
2. If there is an **odd** number of values, the median is the middle number.
3. If there is an **even** number of values, the median is the **average of the two middle numbers**.

Example 1 (Odd Number of Values): Dataset: **3, 5, 7, 9, 11**

- The middle value is **7**, so the median = **7**

Example 2 (Even Number of Values): Dataset: **2, 4, 6, 8**

- The two middle numbers are **4** and **6**
- Median = $(4+6)/2 = 5$

When to Use the Median?

- ✅ Best used when data has **outliers** or is **skewed**, as it is not affected by extreme values.
- ❌ Not ideal for datasets that need precise average calculations.

Correlation

Correlation is a statistical measure that describes the relationship between two variables. It tells us **how strongly** and **in what direction** two variables move together.

Types of Correlation:

1. Positive Correlation

1. When one variable increases, the other also increases.
2. **Example:** More study hours → Higher exam scores.

2. Negative Correlation

1. When one variable increases, the other decreases.
2. **Example:** More screen time → Less sleep.

3. No Correlation —

1. No relationship between the variables.
2. **Example:** Shoe size and intelligence.

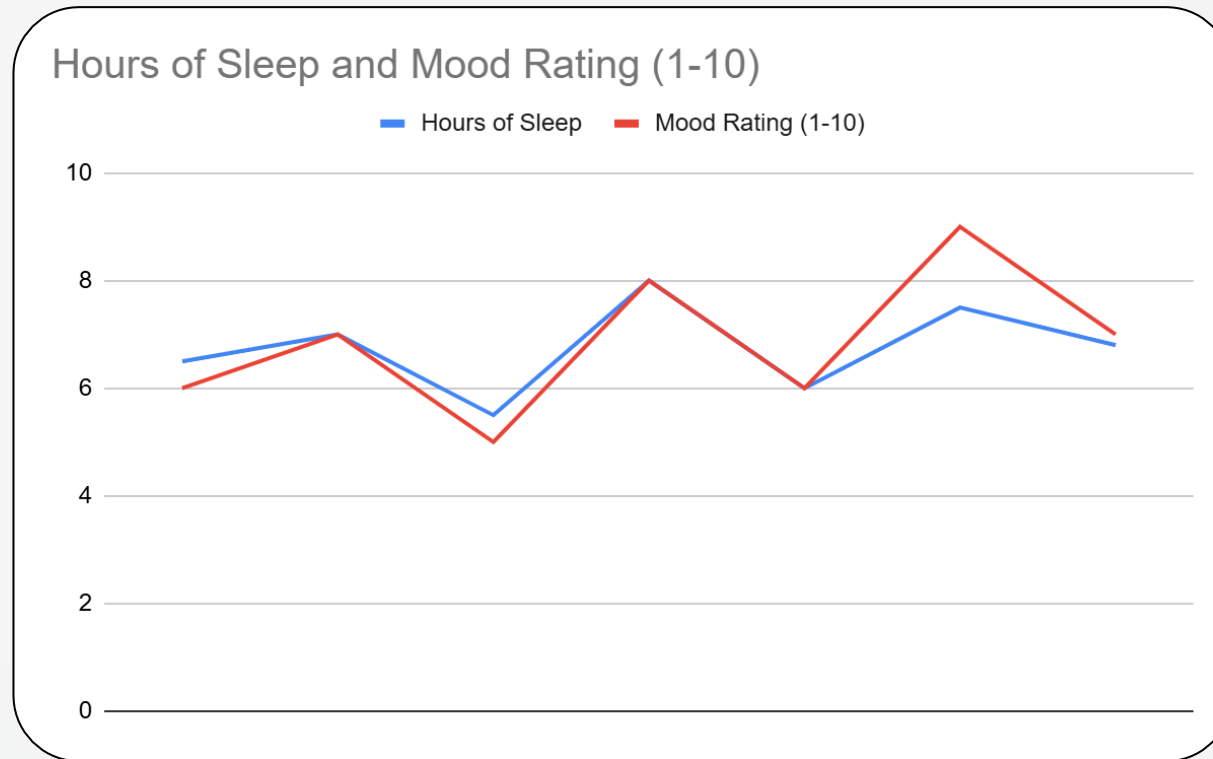
How is Correlation Measured?

The **correlation coefficient (r)** ranges from **-1 to +1**:

- **+1** → Perfect **positive** correlation
- **0** → No correlation
- **-1** → Perfect **negative** correlation

Data Visualization Techniques

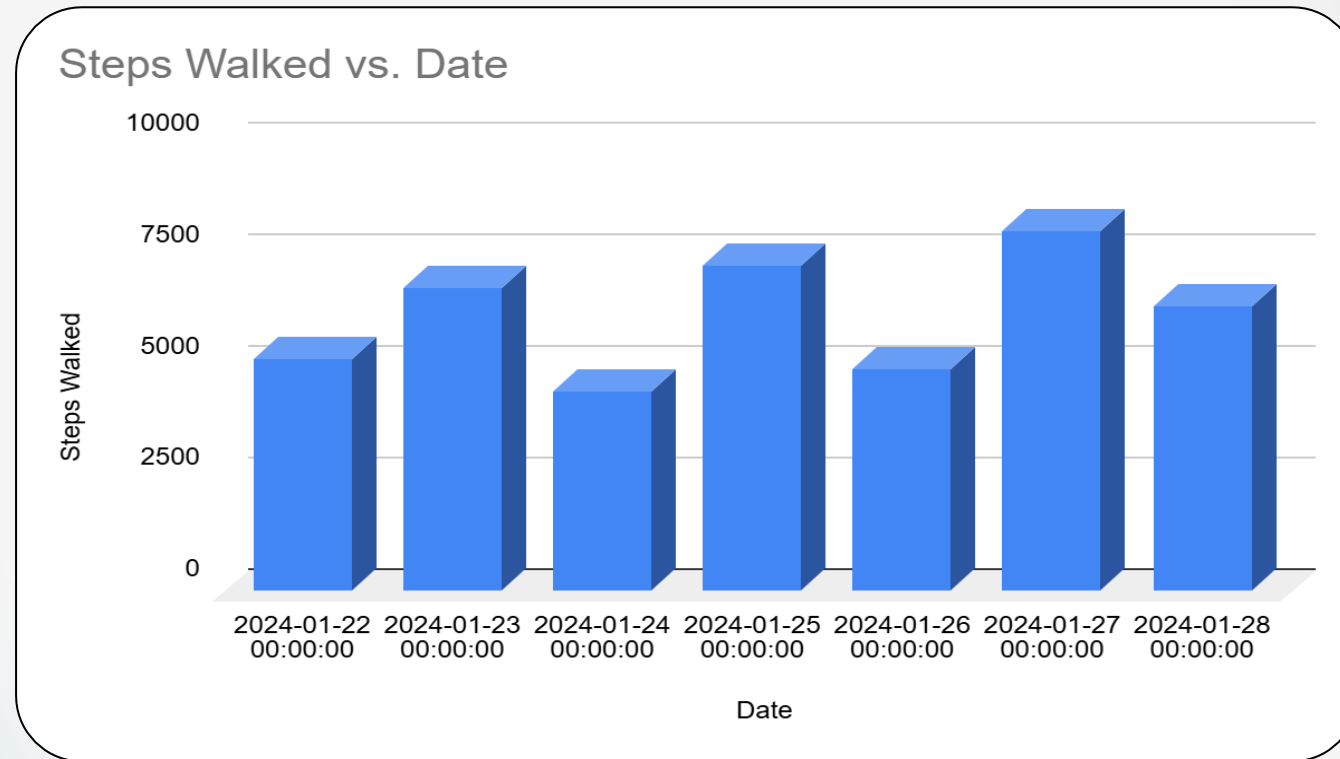
Let's bring the data to life with visualizations



Line Chart: Sleep hours vs. Mood Rating

Data Visualization Techniques

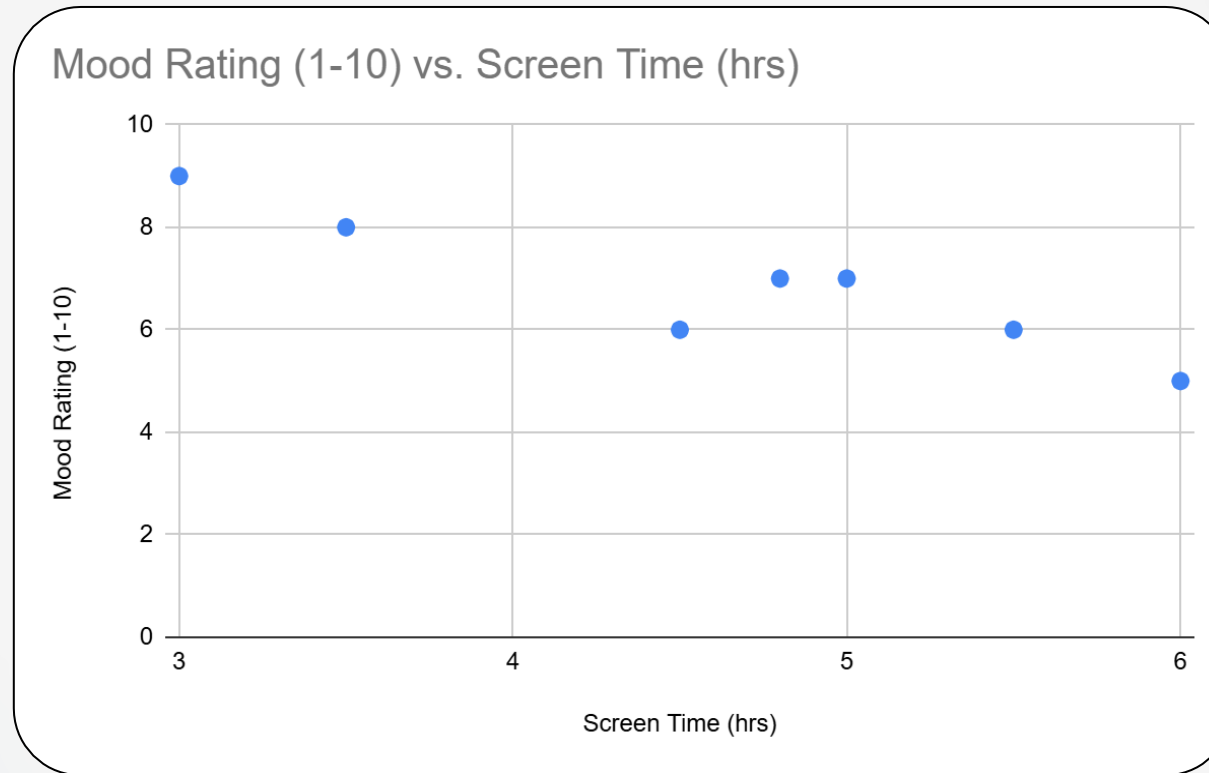
Let's bring the data to life with visualizations



Bar Graph: Daily steps comparison

Data Visualization Techniques

Let's bring the data to life with visualizations



Scatter Plot: Screen time vs. Mood Rating

Hands-On Exercise



Google Sheets

Track your chosen metric (e.g., screen time, steps, sleep) for 1 week.

Input data into Google Sheets.

Create charts and analyze trends.

What You'll Achieve?



Google Sheets

Apply basic statistical concepts to real-life data.

Recognize trends in your daily habits.

Make informed adjustments for a healthier lifestyle.

Knowledge Check :

Q1. What does correlation measure in a dataset?

- A) The total sum of all values in the dataset**
- B) The relationship between two variables**
- C) The difference between the highest and lowest values**
- D) The average value of a dataset**

Knowledge Check :

Q1. What does correlation measure in a dataset?

- A) The total sum of all values in the dataset**
 - B) The relationship between two variables**
 - C) The difference between the highest and lowest values**
 - D) The average value of a dataset**
- ✓ Correct Answer: B) The relationship between two variables**

Knowledge Check :

Q2. When is the median a better measure of central tendency than the mean?

- A) When the data has no outliers**
- B) When all values are the same**
- C) When the data is skewed or has extreme values**
- D) When calculating the total sum of all values**

Knowledge Check :

Q2. When is the median a better measure of central tendency than the mean?

- A) When the data has no outliers**
- B) When all values are the same**
- C) When the data is skewed or has extreme values**
- D) When calculating the total sum of all values**

 **Correct Answer: C) When the data is skewed or has extreme values**

Knowledge Check :

Q3. What type of data is best visualized using a bar chart?

- A) Continuous data like temperature trends**
- B) Categorical data like sales of different product categories**
- C) Relationship between two numerical variables**
- D) Variability in a dataset over time**

Knowledge Check :

Q3. What type of data is best visualized using a bar chart?

- A) Continuous data like temperature trends**
- B) Categorical data like sales of different product categories**
- C) Relationship between two numerical variables**
- D) Variability in a dataset over time**

 **Correct Answer: B) Categorical data like sales of different product categories**

Knowledge Check :

Q4. What is the primary purpose of a line chart?

- A) To compare different categories**
- B) To show trends over time**
- C) To display relationships between two variables**
- D) To represent frequencies of different categories**

Knowledge Check :

Q4. What is the primary purpose of a line chart?

- A) To compare different categories**
- B) To show trends over time**
- C) To display relationships between two variables**
- D) To represent frequencies of different categories**

 **Correct Answer: B) To show trends over time**

Knowledge Check :

Q5. A scatter plot is most useful for which type of analysis?

- A) Showing the distribution of data within categories**
- B) Identifying the relationship between two numerical variables**
- C) Comparing multiple categories using bars**
- D) Summarizing large amounts of data into a single value**

Knowledge Check :

Q5. A scatter plot is most useful for which type of analysis?

- A) Showing the distribution of data within categories**
- B) Identifying the relationship between two numerical variables**
- C) Comparing multiple categories using bars**
- D) Summarizing large amounts of data into a single value**

 **Correct Answer: B) Identifying the relationship between two numerical variables**

Knowledge Check :

Q6. What is the mean of a dataset?

- A) The middle value when the data is arranged in order
- B) The most frequently occurring value in a dataset
- C) The sum of all values divided by the number of values
- D) The difference between the highest and lowest values

Knowledge Check :

Q6. What is the mean of a dataset?

- A) The middle value when the data is arranged in order
- B) The most frequently occurring value in a dataset
- C) The sum of all values divided by the number of values
- D) The difference between the highest and lowest values

 **Correct Answer: C) The sum of all values divided by the number of values**

Knowledge Check :

Q7. What is the median in a dataset?

- A) The middle value when the data is arranged in ascending or descending order
- B) The average of all values in the dataset
- C) The most frequently occurring value
- D) The largest value minus the smallest value

Knowledge Check :

Q7. What is the median in a dataset?

- A) The middle value when the data is arranged in ascending or descending order
- B) The average of all values in the dataset
- C) The most frequently occurring value
- D) The largest value minus the smallest value

 **Correct Answer: A) The middle value when the data is arranged in ascending or descending order**

Summary

- Factors contributing in the better mood ratings are
 - a) Hours of sleep
 - b) Steps Walked
 - c) Water Intake
- Factors contributing in the lower mood ratings is Screen time .
- Key take ways :
 - Reduce the screen time for better mood.
 - walk more, drink plenty of water to stay hydrated and happy
 - Have enough hours of sleep for better mood during day.